Lab 2: Install and configure Bind9 so that the client can access the website in Lab1 with a domain name

Tags

Part 1: Configure on Web server

Step 1: Install Bind9 on Web Server

`sudo` `apt update`

The `sudo` play a role as administrator privileges when running the command, and apt update is refresh the system's package list from online repositories

`sudo` `apt install bind9 dnsutils -y`

The apt install is used for install new software packages which is bind9, the DNS server. The `dnsutils` is tools like `dig`, `nslookup`, and `host` used for testing.

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update
[sudo] password for kali:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
235 packages can be upgraded. Run 'apt list --upgradable' to see them.

┌──(kali㉿kali)-[~]
└─$ sudo apt install bind9 dnsutils -y
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
bind9-dnsutils is already the newest version (1:9.20.9-1).
Installing:
  bind9

Suggested packages:
  bind9-doc  resolvconf

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 235
  Download size: 250 kB
  Space needed: 909 kB / 64.2 GB available

Get:1 http://mirror.aktkn.sg/kali kali-rolling/main amd64 bind9 amd64 1:9.20.9-1 [250 kB]
Fetched 250 kB in 1s (266 kB/s)
Selecting previously unselected package bind9.
(Reading database ... 410992 files and directories currently installed.)
Preparing to unpack .../bind9_1%3a9.20.9-1_amd64.deb ...
Unpacking bind9 (1:9.20.9-1) ...
Setting up bind9 (1:9.20.9-1) ...
wrote key file "/etc/bind/rndc.key"
update-rc.d: We have no instructions for the named init script.
update-rc.d: It looks like a network service, we disable it.
named-resolvconf.service is a disabled or a static unit, not starting it.
named.service is a disabled or a static unit, not starting it.
Processing triggers for ufw (0.36.2-9) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.3.0) ...
```

Step 2: Define the Domain Zone in Bind9

`sudo nano /etc/bind/named.conf.local`

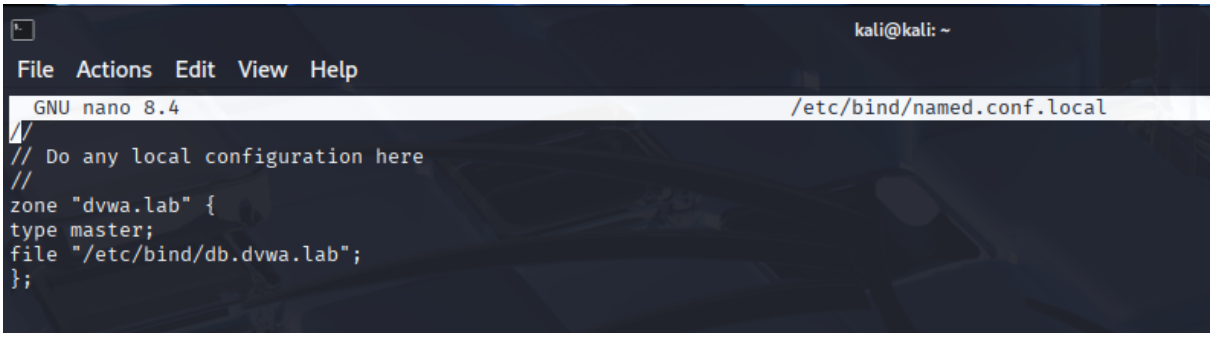
The nano is a command to open a text editor and `/etc/bind/named.conf.local` is where the file of DNS zones is declared.

```
zone "dvwa.lab" {
type master;
file "/etc/bind/db.dvwa.lab";
};
```

`"dvwa.lab"` defines the custom DNS domain.

`type master` indicates that this server is authoritative.

`file` tells Bind9 where to load the zone records from.



Step 3: Create a zone File:

`sudo nano /etc/bind/db.dvwa.lab`

Creates a zone file that maps hostnames to IP addresses.
Contents of the file:

```
$TTL    604800
@       IN      SOA     ns.dvwa.lab. admin.dvwa.lab. (
                2024071901 ; Serial
                604800     ; Refresh
                86400      ; Retry
                2419200    ; Expire
                604800 )   ; Negative Cache TTL

@       IN      NS      ns.dvwa.lab.
```

```
ns      IN      A       192.168.88.129
@       IN      A       192.168.88.129
dvwa    IN      A       192.168.88.129
www     IN      A       192.168.88.129
db      IN      A       192.168.88.130
```

`$TTL 604800` : Time to cache DNS responses (7 days).

`@` : Represents the root of the zone, `dvwa.lab` .

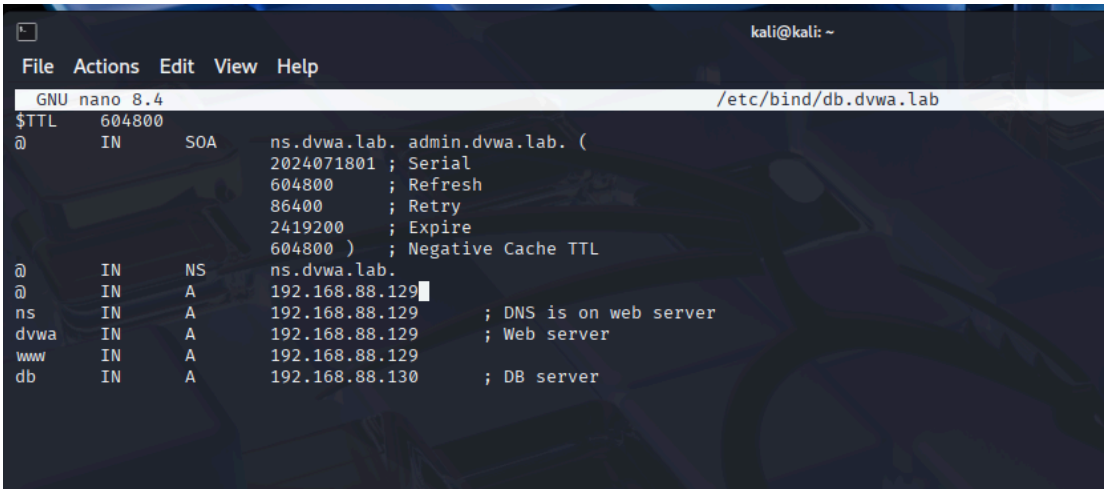
`IN` : Internet class.

`SOA` : Start of Authority, identifies main DNS server and contact.

`NS` : Declares name server for the zone.

`A` : Maps hostnames to IPs.

- `ns` , `dvwa` , `www` → Web Server
- `db` → Database Server



Step 4: Validate and Restart DNS Server

`sudo named-checkzone dvwa.lab /etc/bind/db.dvwa.lab`

Checks syntax of the zone file.

`sudo systemctl restart named`

Restarts Bind9 to apply the new configuration. On Kali, the service name is `named` .

```
┌──(kali㉿kali)-[~]
└─$ sudo named-checkconf

┌──(kali㉿kali)-[~]
└─$ sudo named-checkzone dvwa.lab /etc/bind/db.dvwa.lab
zone dvwa.lab/IN: loaded serial 2024071801
OK

┌──(kali㉿kali)-[~]
└─$ sudo systemctl restart named

┌──(kali㉿kali)-[~]
└─$
```

Step 5: Configure DNS Resolver Locally

`sudo nano /etc/resolv.conf`

This file tells the system which DNS server to query.

Contents:

```
nameserver 192.168.88.129
```

```
kali@kali: ~
File  Actions  Edit  View  Help
  GNU nano 8.4                                    /etc/resolv.conf
# Generated by NetworkManager
search localdomain
nameserver 192.168.88.129
search dvwa.lab
```

Step 6: Test DNS Resolution on Web Server

`dig dvwa.lab`
`dig db.dvwa.lab`
`ping dvwa.lab`
`ping db.dvwa.lab`

Expected responses include IP addresses from the zone file, confirming DNS resolution works.Z

```
┌──(kali㉿kali)-[~]
└─$ dig dvwa.lab

; <<>> DiG 9.20.9-1-Debian <<>> dvwa.lab
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 46848
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: cf478ed1b1d0cece01000000687787512 8e4e9ecd4797d2e (good)
;; QUESTION SECTION:
;dvwa.lab.                      IN      A

;; ANSWER SECTION:
dvwa.lab.               604800  IN      A       192.168.88.129

;; Query time: 0 msec
;; SERVER: 192.168.88.129#53(192.168.88.129) (UDP)
;; WHEN: Wed Jul 16 07:04:49 EDT 2025
;; MSG SIZE  rcvd: 81


┌──(kali㉿kali)-[~]
└─$ ping dvwa.lab
PING dvwa.lab (192.168.88.129) 56(84) bytes of data.
64 bytes from 192.168.88.129: icmp_seq=1 ttl=64 time=0.086 ms
64 bytes from 192.168.88.129: icmp_seq=2 ttl=64 time=0.066 ms
^Z
zsh: suspended  ping dvwa.lab

┌──(kali㉿kali)-[~]
└─$ ping db.dvwa.lab
PING db.dvwa.lab (192.168.88.130) 56(84) bytes of data.
64 bytes from 192.168.88.130: icmp_seq=1 ttl=64 time=1.46 ms
64 bytes from 192.168.88.130: icmp_seq=2 ttl=64 time=4.07 ms
64 bytes from 192.168.88.130: icmp_seq=3 ttl=64 time=1.68 ms
64 bytes from 192.168.88.130: icmp_seq=4 ttl=64 time=1.59 ms
64 bytes from 192.168.88.130: icmp_seq=5 ttl=64 time=2.86 ms
^Z
zsh: suspended  ping db.dvwa.lab

┌──(kali㉿kali)-[~]
└─$
```

Part 2: Configure on Database server

Step 1: Point DNS to the Web Server

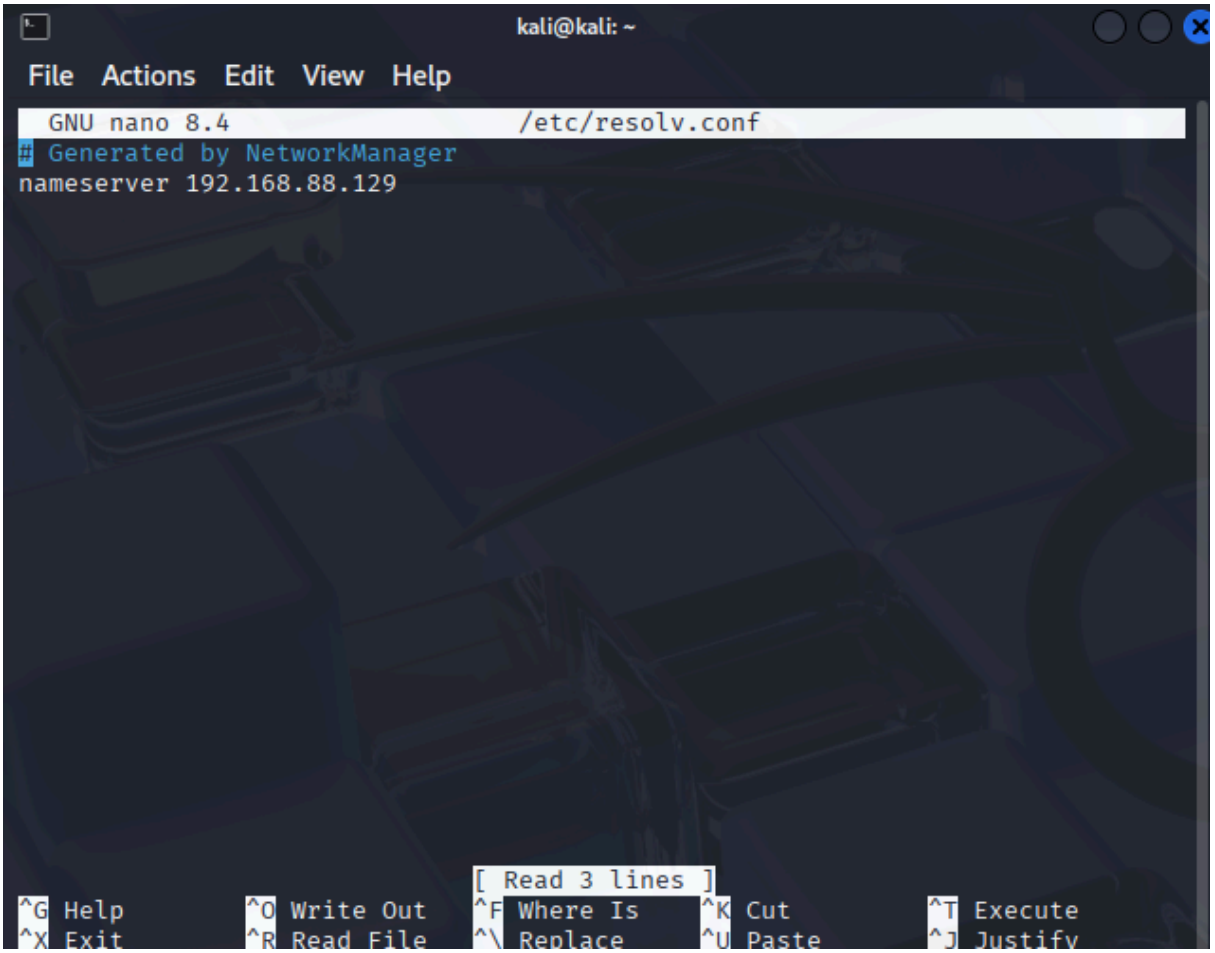
`sudo nano /etc/resolv.conf`

Edit the file to define the DNS nameserver.

Set this:

`nameserver 192.168.88.129`

This tells the DB Server to query the Web Server's Bind9 for hostname resolution.



Step 2: Test DNS from DB Server

bash

```
dig dvwa.lab
dig db.dvwa.lab
ping dvwa.lab
ping db.dvwa.lab
```

Successful results confirm the DB Server can resolve the domain names defined in Bind9.

```
└$ sudo nano /etc/resolv.conf


┌──(kali㉿kali)-[~]
└─$ ping dvwa.lab
PING dvwa.lab (192.168.88.129) 56(84) bytes of data.
64 bytes from 192.168.88.129: icmp_seq=1 ttl=64 time=0.557 ms
64 bytes from 192.168.88.129: icmp_seq=2 ttl=64 time=1.48 ms
64 bytes from 192.168.88.129: icmp_seq=3 ttl=64 time=1.13 ms
64 bytes from 192.168.88.129: icmp_seq=4 ttl=64 time=1.54 ms
^Z
zsh: suspended  ping dvwa.lab

┌──(kali㉿kali)-[~]
└─$ dig db.dvwa.lab

; <<>> DiG 9.20.9-1-Debian <<>> db.dvwa.lab
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 20072
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: a8d43475b375b311010000068778a7379b9577ca867c4da (good)
;; QUESTION SECTION:
;db.dvwa.lab.                   IN      A

;; ANSWER SECTION:
db.dvwa.lab.            604800  IN      A       192.168.88.130

;; Query time: 0 msec
;; SERVER: 192.168.88.129#53(192.168.88.129) (UDP)
;; WHEN: Wed Jul 16 07:18:11 EDT 2025
;; MSG SIZE  rcvd: 84
```

```
┌──(kali㉿kali)-[~]
└─$ ping db.dvwa.lab
PING db.dvwa.lab (192.168.88.130) 56(84) bytes of data.
64 bytes from 192.168.88.130: icmp_seq=1 ttl=64 time=0.110 ms
64 bytes from 192.168.88.130: icmp_seq=2 ttl=64 time=0.051 ms
64 bytes from 192.168.88.130: icmp_seq=3 ttl=64 time=0.049 ms
^Z
zsh: suspended  ping db.dvwa.lab

┌──(kali㉿kali)-[~]
└─$ dig dvwa.lab

; <<>> DiG 9.20.9-1-Debian <<>> dvwa.lab
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 60404
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 730b043dc39bd64c0100000068778abb889b970555920872 (good)
;; QUESTION SECTION:
;dvwa.lab.                      IN      A

;; ANSWER SECTION:
dvwa.lab.               604800  IN      A       192.168.88.129

;; Query time: 3 msec
;; SERVER: 192.168.88.129#53(192.168.88.129) (UDP)
;; WHEN: Wed Jul 16 07:19:24 EDT 2025
;; MSG SIZE  rcvd: 81
```

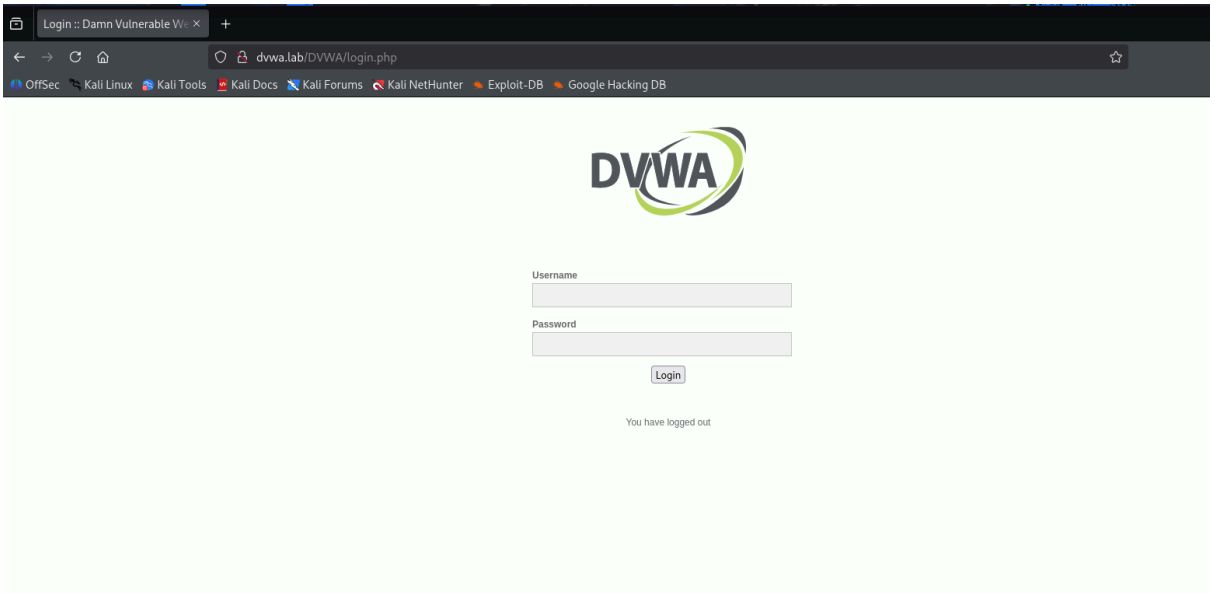
Part 3: Access website by domain name

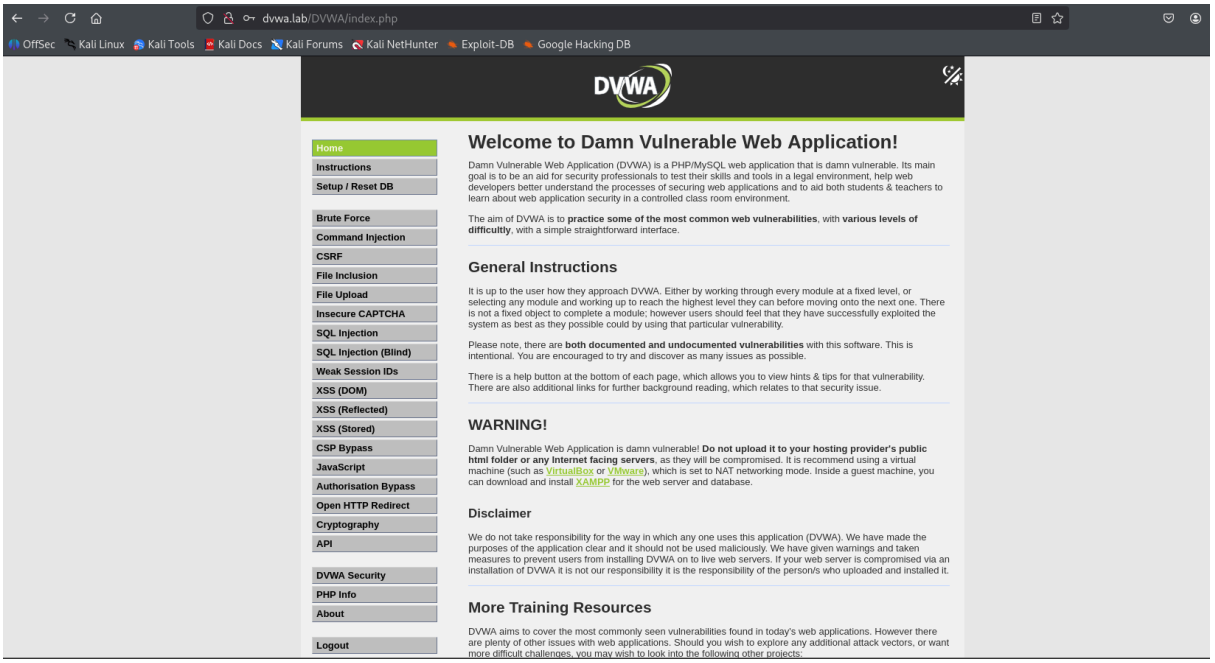
Step 10: Web access via DNS

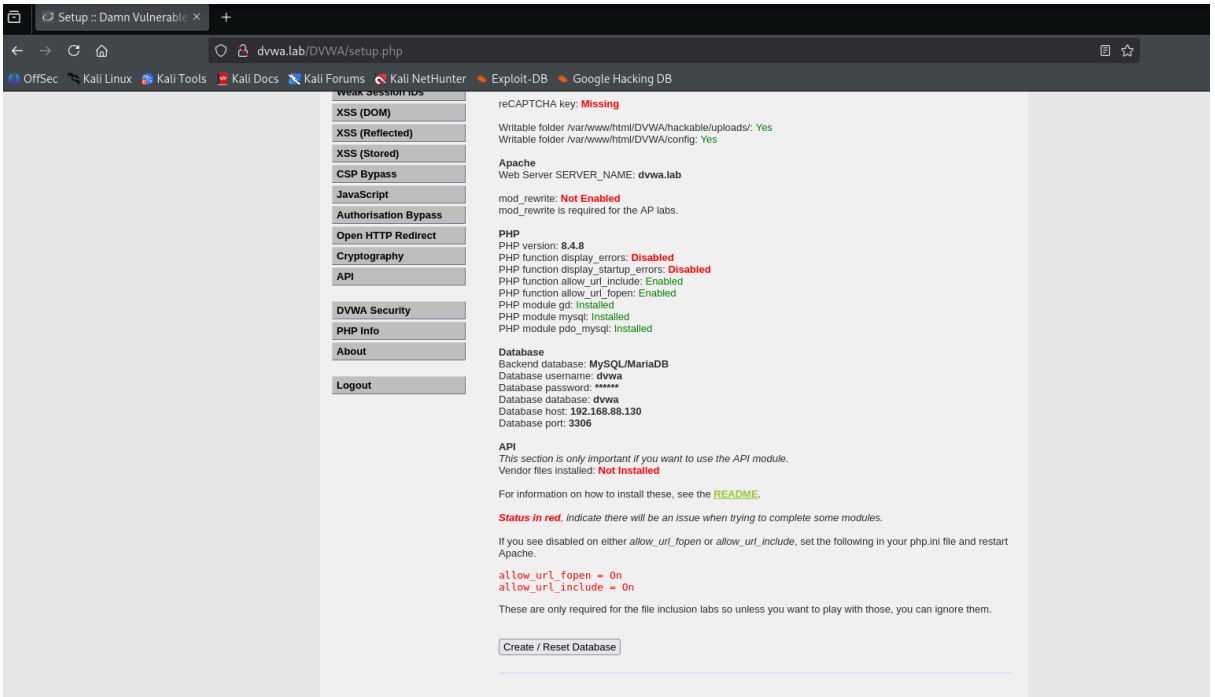
Opens a browser and enters:

http://dvwa.lab/DVWA/setup.php

And by logging in using username: admin and password: password






Setup :: Damn Vulnerable
dvwa.lab/DVWA/setup.php
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
Authorisation Bypass
Open HTTP Redirect
Cryptography
API
DVWA Security
PHP Info
About
Logout
reCAPTCHA key: Missing
Writable folder /var/www/html/DVWA/hackable/uploads/: Yes
Writable folder /var/www/html/DVWA/config: Yes
Apache
Web Server SERVER_NAME: dvwa.lab
mod_rewrite: Not Enabled
mod_rewrite is required for the AP labs.
PHP
PHP version: 8.4.8
PHP function display_errors: Disabled
PHP function display_startup_errors: Disabled
PHP function allow_url_include: Enabled
PHP function allow_url_fopen: Enabled
PHP module gd: Installed
PHP module mysql: Installed
PHP module pdo_mysql: Installed
Database
Backend database: MySQL/MariaDB
Database username: dvwa
Database password: ******
Database database: dvwa
Database host: 192.168.88.130
Database port: 3306
API
This section is only important if you want to use the API module.
Vendor files installed: Not Installed
For information on how to install these, see the README.
Status in red, indicate there will be an issue when trying to complete some modules.
If you see disabled on either allow_url_fopen or allow_url_include, set the following in your php.ini file and restart Apache.
allow_url_fopen = On
allow_url_include = On
These are only required for the file inclusion labs so unless you want to play with those, you can ignore them.
Create / Reset Database